

## **The Future of Consumer Transportation: A Vision of Cars and Traffic by 2060**

### ***Introduction:***

By 2060, the way people move will look drastically different from the way we do today. This is largely due to technological advancements in autonomous vehicles, which are systematically changing the way society functions. In this paper, we will consider a “Vision for 2060” through the following lenses: Vehicle Design, Congestion Management, Social and Economic Factors, Energy and Sustainability, and Ethical Implications.

### ***The Future of Vehicle Design:***

The design of motor vehicles in 2060 will be radically different from the most advanced vehicles of today. Today’s most intelligent autonomous vehicles, Waymos and Teslas, are already quite advanced. They already operate with many advanced hardware and software systems, including a cutting edge suite of sensing technology and Vision Language Models that take inspiration from recent innovations in LLMs and deep learning to make accurate driving decisions faster than the blink of a human eye (Tian et al). They are, however, still retrofitted manual drive vehicles: they have steering wheels, front facing seats, and side-view mirrors, vestiges designed for human drivers that are rarely there (Waymo). These features take up extra space and contribute needlessly to the cost of each vehicle (Peña). Fleet managers will need to buy and maintain autonomous fleets that are orders of magnitude larger than those of today to keep pace with rideshare demand. If these vestige-ridden modern designs are scaled up to fleets of tens of thousands of vehicles, they will be woefully inefficient.

### Optimizing Design for Cost:

Existing visions for the near future already have these removed, for example the emergent Zoox cab. The first of its kind, this cab will look more like a gondola on wheels than a modern car. Lacking any of the aforementioned vestiges, it will instead boast floor-to-ceiling doors and inward facing passenger benches, offering customers a much more spacious ride. Additionally, it will have the ability to drive in either direction and perform four-wheel steering maneuvers, significantly improving its access for pickups on congested urban roads (Zoox). These vehicles will not only provide riders with a better experience, they will also reduce fleet costs significantly.

As we move further into the future, more opportunities will present themselves for cost reduction. For example, batteries may improve in efficiency but will remain a significant cost and weight on the vehicle. Future autonomous vehicles will look for an alternative source of power, relying on a constant, direct link to the grid via a technology called Dynamic Wireless Power Transfer (DWPT). DWPT uses small copper coils either placed on top of or ingrained directly into the road. These coils are connected to the grid, and through induction will charge vehicles while they drive as needed. Vehicles will have significantly less need for batteries, especially as they are used increasingly in urban and suburban areas where rollout will likely occur first. Not only will the reduction in battery capacity reduce the cost and weight of batteries directly, it will also require much less robust braking and suspension systems due to the reduced weight and significantly increase the efficiency of the vehicle. (Electreon) AVs will become much, much cheaper.

### Optimizing Design for Riders and Pedestrians:

The development of a fleet-subscription system will also yield a convergence in marginal costs between rides in mass market and luxury vehicles. As ridership increases and business models transition from selling individual vehicles sporadically to subscriptions reliably, the cost of maintaining each vehicle in a luxury fleet can be spread out over a vehicle's lifetime mileage. Waymo, for example, already sees 70% of the miles its vehicles drives filled with paying customers (Watson, Tran, and Clarke). This will mean that the subscription costs for luxury fleets will only need to be marginally more expensive than those for mass market fleets to remain profitable. However, we do not predict that the convergence in production costs will result in a convergence in subscription price. Like other luxury goods (also known as Veblen goods), charging a higher price will actually increase demand for it from first-class clients, who will perceive it as a high-quality good. Thus, there will be a divergence in the market for autonomous vehicle subscriptions between mass-market providers who seek to compete on cost and premium services that compete on quality.

This market split will result in a bifurcation in fleet designs. Though both will maintain many common elements, like inward facing seats and floor-to ceiling doors, there will be significant differences. Mass market vehicles will likely use cheaper materials for seating and interior finishes, and they will be less likely to have as many or as sophisticated customer comfort features as luxury vehicles do. Luxury vehicles, in an effort to differentiate themselves enough to justify their prices and make their clients feel elite, will tend to use more expensive materials such as mahogany for finishes and leather for seating. They will also have customer interface features that make passenger-vehicle interactions seamless: voice commands, haptic controls, and eye tracking will all be possible (Mercedes-Benz).

Vehicles will not only be able to better communicate with passengers to facilitate comfort, they will also need to be able to communicate with pedestrians. In modern driving systems, pedestrians deduce the next actions of a vehicle by making eye contact with drivers, using hand signals, calling out the window, or using blinkers. The driverless vehicles of the future will not have access to these features. Instead, according to Mercedes, they will use laser projection and LLM-assisted speakers to communicate with pedestrians. They will be able to give basic instructions (and perhaps hold longer conversations with LLM assistance), project crosswalks, and indicate exactly where and when it intends to park. With the dual impetuses of OEMs seeking consumer trust and regional governments seeking effective safety regulation, pedestrians will experience a dramatic increase in roadside safety and agency (Mercedes-Benz).

The autonomous vehicles of the future will experience a significant evolution of design to reflect different market priorities. As vehicle ownership transitions from individuals to large scale fleet managers, a higher prioritization of vehicle costs will motivate companies to remove unnecessary vestiges of manual drivers and adopt alternative power sources. In addition, a rise in subscription-based business models will allow fleet managers to amortize vehicle costs over time. This will lead to a much smaller difference in the cost of providing a luxury and mass-market fleet, will lead to a bifurcation in the market, and will yield differentiated designs for the two different markets. Finally, a combination of government pressure and providers' need to earn rider trust will motivate AV's to design for safety in exciting new ways. Though the cars of the future will look quite different than the AVs of today, they will make advances that provide customers with a safer, cheaper, and better ride experience.

***Congestion Management and V2V Communication:***

Although this futuristic vision for cars sounds exciting, the defining feature of transportation will not be whether vehicles can drive themselves, but whether they can collaborate effectively. Congestion management in a future dominated by AVs will depend upon how successfully a network of vehicles and infrastructures can coordinate. Smart city transit will draw from advances in peer-to-peer networking and incentives encouraging interoperability between manufacturers.

Effective congestion management in 2060 rests on the concept of Vehicle-to-Everything (V2X) communication, which encompasses data exchange among vehicles (V2V) and infrastructure such as traffic systems (V2I). Dedicated Short-Range Communications (DSRC) has been used and Cellular V2X (C-V2X), has recently emerged by utilizing 4G/5G networks to enhance data exchange and lower latency (Yusuf et. al, 2024). As a result, C-V2X enables collaborative perception which allows vehicles to share sensor data to foresee situations beyond their own line of sight.

However, there are many barriers to making this vision a reality. Some include economic viability with level 4/5 adoption, , debates over architecture, and countless legal challenges. A Red Hat engineer emphasized that only 5% of vehicles will be at Level 4 or 5 by 2035, citing thermal management as the key hidden constraint beyond economic viability. Right now, powertrains dominate vehicle energy demands, but cooling dense compute stacks will limit production and adoption of L4/L5 vehicles (personal communication, February 2026). A Waymo engineer also stressed that geographic mapping and city integration/testing presents a barrier to scaling, despite the increasing amount of training data becoming available (personal communication, March 2026). In addition, ecosystem fragmentation must be addressed before V2V can scale. Waymo might be able to coordinate internally, but cannot communicate with

Tesla vehicles for lane merges or braking. For autonomous vehicles to scale rapidly in society, vehicles from different manufacturers must be able to communicate seamlessly and collaborate effectively.

#### Platooning and Traffic:

One of the most promising applications of V2V is platooning, where tightly-spaced vehicles can travel synchronously. Each car shares precise position, acceleration, and braking telemetry with the others, allowing for minimal following distances without human reaction delays. Benefits include increased lane capacity and reduced energy consumption due to aerodynamic efficiency.

Platooning, however, requires very low latency and incredibly secure cyber infrastructure. A brief communication lag or a hack could destabilize hundreds of vehicles in milliseconds. Hybrid roads with both AVs and humans further complicate the rollout of platooning. Nevertheless, platooning is projected to reduce both the frequency and severity of accidents. According to an engineering director at General Motors, platooning has the potential to reduce accident severity (personal communication, February 2026). This is largely due to coordinated braking systems that prevent sudden speed gaps between vehicles. This results in a lower  $\Delta v$  (change in velocity), as being closer together reduces both collision energy and the force of the crash, thus reducing accident severity.

#### Smart Optimization and Public Integration:

The potential for smart optimization is endless. Through better clearing and faster rerouting, V2X coordination can cut travel times by 15–18% and speed emergency response times by up to 20–25%. Environmentally trained algorithms have the potential to reduce CO<sub>2</sub> by 10–15% and mitigate idle fuel waste through platooning (Kumar & Mohd, 2025). The effects of

V2V networking will allow for greater emission reduction, optimal energy usage, and traffic flow that humans could never manually coordinate.

Although widespread adoption is imminent, city integration remains a challenge. To address this, engineers have considered optimizing public transportation demand. Instead of operating 50-person empty late-night buses, municipal systems could dispatch autonomous shuttles or cars on an on-demand schedule integrated into the same cloud network as private vehicles. This system would save energy and money, reduce low public transit utilization rates, and provide accessible service in low-demand periods, all while complimenting existing transit systems.

#### Network Integration Across Manufacturers:

System-level congestion reduction will remain limited until all manufacturers and platforms cooperate through shared digital infrastructure. For example, product managers at May Mobility said that companies like Uber and Lyft are integrating autonomous vehicles into their offerings (personal communication, February 2026). In fact, third party network providers like Uber and Lyft could link multiple different AV providers under their umbrella network. However, Cross-OEM integration will require policy-driven incentives and government-enforced standards, similar to how air traffic controllers manage flights for different airlines at the same airport.

Going forward, the future of widespread autonomous vehicles will rely on a centralized system that has V2V and V2I capabilities. This will serve as the foundation for a futuristic society where autonomous vehicles are pervasive and congestion management is fundamentally altered and seamlessly integrated.

## ***Societal and Economic Impact***

### **Consumers:**

As mentioned previously, today's system of individual car ownership will eventually be replaced by monthly subscriptions to autonomous ridesharing companies. These monthly subscription costs will range from the low three digits to high four digits, depending on usage and vehicle luxuriousness. The reason for this inevitable development lies in the economics of owning a car. A study from Germany found that "passenger-car fleet[s] [are] around 50 million vehicles, which could potentially provide 250 million seats" (RAFA1). Since private cars are used only 5% of the day and carry an average of just one to two passengers at a time, these vehicles operate at a mere 2% of its total seat capacity (RAFA1). As a result, cars are one of the most expensive and inefficiently used consumer goods in the world. In fact, a study from McKinsey Center for Future Mobility (MCFM) found that this inefficiency will promote enormous growth in the shared mobility market. The same study found that the ridesharing market is predicted to grow to \$1 Trillion in 2030, fueled by the rapid expansion of AV-taxi fleets like Waymo. Initially, densely populated urban areas will experience most of this growth, while suburbs and rural areas are likely to maintain privately owned vehicles with advanced automation and connectivity.

In a 2025 interview with Uber CEO, Dara Khosrowshahi, Khosrowshahi predicted that enormous advances in AV's over the next 20 years will make human driving "as niche a skill as horseback riding is today" (RAFA2). This quote directly coincides with our predictions that autonomous ride sharing subscriptions will make driver's licenses obsolete. As a replacement, the government will likely issue a limited and expensive supply of licenses that will shift driving to a form of recreation, leisure, and enjoyment. As far as insurance goes, one of the largest firms,



Swiss RE, expects that “insurance will be marketed to owners of car fleets rather than individuals,” and liability will shift from the user to car manufacturers/designers (RAFA3).

As a result, self-driving cars will be much safer than human driving. This will likely manifest in the form of exploding insurance costs for human drivers. Furthermore, human driving will become a luxury hobby like horseback riding is today.

As far as safety goes, Waymo and Swiss RE reported that across 25.3 million fully autonomous miles, Waymo’s system resulted in an 88% reduction in property-damage claims and a 92% reduction in bodily-injury claims relative to human drivers (RAFA4). These numbers will improve until accident rates approach zero. As a result of these massive safety improvements, experts estimate that the U.S. organ supply could fall about 14–18%. Coupled with already long waitlists for donations, autonomous driving could lead to an unprecedented organ shortage.

#### Car Manufacturers:

Since consumers will pivot away from owning cars, car manufacturers will dramatically change their current business models. Many car manufacturers will become global autonomous ride-sharing companies that control the entire value creation pipeline, including building the cars, operating the AV ridesharing fleets, and licensing their autonomous driving software (RAFA6). Although most ride-hailing companies will struggle with profitability, the potential elimination of drivers by 2050 will also eliminate their largest expense: humans (RAFA6). However, in order to reach profitability comparable to competition like internal combustion engines (ICE), car manufacturers must pursue adjacent value creation activities. For example, electric AV fleets will generate massive revenue from vehicle-to-grid and electricity arbitrage where cars will essentially be batteries on wheels. As a result, these cars can give back to the grid during peak demand events or electricity outages, making autonomous EV fleets material actors in the future

grid (RAFA8). By 2035, this adjacent energy value pool for passenger cars will surpass \$400B (RAFA7).

The impacts on single car manufacturers explained above will also change the structure of the entire auto industry. The western auto industries which traditionally profited from mechanical engineering and manufacturing are currently experiencing heightened competition from Asia where cars are now software-defined products. In order to remain competitive, experts also predict that manufacturers will have to invest around \$100B for AV development and EV platform transition (RAFA10). These developments will likely lead to an unprecedented industry consolidation, and it is likely that car brands that have been household names for over a century will either disappear or unrecognisably change in the next two decades.

### ***Energy and Sustainability***

Replacing privately owned cars with electric AV fleets will demand a fundamental rethinking of how cities are planned, how electricity is distributed, and who will benefit from the transition. While EV adoption is accelerating globally, the technology carries with it a complex set of tradeoffs that urban planners, policymakers, and consumers are only beginning to fully reckon with.

A PRISMA systematic review conducted by Tilly et al. (2024) maps the full landscape of externalities accompanying mass EV adoption that the future of autonomy relies upon. On the environmental side, EV battery production is extraordinarily resource-intensive. The manufacturing phase alone accounts for over 65% of an EV's direct material footprint, and is driven by demand for lithium, cobalt, nickel, and copper. As Tilly et al. note, “the EV market alone will require 26 times more lithium, 6 times more cobalt, 12 times more nickel, and nine

times more graphite between 2021 and 2050” (Tilly et al., 2024). These extraction pressures create serious biodiversity and water scarcity concerns.

#### Environmental Tradeoffs:

The electricity used to charge EVs is still predominantly sourced from fossil fuel, complicating the clean energy narrative. However, advances in vehicle-to-grid technology present a meaningful countervailing opportunity: EV batteries retain 70-80% of their original charge capacity even after their automotive lifespan, meaning retired battery packs could feasibly satisfy approximately 30% of global demands for grid-based energy storage by 2030. This points toward a future in which EV fleets could be active participants in stabilizing renewable energy grids, storing excess solar and wind power during periods of low demand and feeding it back during peak hours. Ultimately, the long-term sustainability of EV adoption depends heavily on the pace of renewable energy expansion alongside it. Tilly et al. ultimately call for a wider policy framework, one that includes public transit expansion and active transportation investment, rather than assuming that one-to-one replacement of combustion vehicles with EVs is sufficient on its own.

#### Urban Charging Infrastructure:

This challenge is exactly what the urban planning literature is beginning to address through increasingly sophisticated frameworks. Zhang and Kontou (2025) develop a multi-criteria decision-making approach for deploying curbside EV charging stations in San Francisco, noting that “curbside charging has emerged as a practical solution, especially in dense urban areas with limited off-street parking” (Zhang and Kontou, 2025). This matters because a large share of urban residents live in multi-unit dwellings without access to home charging; and in New York, over 60% of residential units are multifamily housing.

The criteria considered in citing decisions reflect the genuine complexity of urban charging deployment, involving economic factors like commuter demand, residential parking pressure, and commercial vehicle activity. Some fairness criteria introduce a need for proximity to healthcare facilities and universities, whose staff often rely on personal vehicles during off-peak hours. Other criteria discuss spatial distribution of existing stations to avoid redundancy and ensure underserved areas are reached. Environmental criteria incorporate flood zone risk and air quality data, directing emissions reduction benefits toward communities with the poorest baseline conditions. The interaction between these criteria is rarely straightforward, as areas with the highest commercial parking demand frequently overlap with neighborhoods that already have the best existing charging coverage. This means a purely demand-driven approach may reinforce rather than correct existing disparities.

Thus, to assess which deployment strategy produces the most value, Zhang and Kontou incorporate a benefit evaluation system that categorizes outcomes across environmental, economic, and social dimensions. Environmental benefits are estimated by calculating emissions reductions achieved by replacing conventional vehicle trips with EV trips. Economic benefits are quantified through fuel cost savings per mile under local utility pricing. The social dimension is the most nuanced, relying on statistics and bivariate mapping to assess whether accessibility improvements are landing in communities that actually need them most. The “high need” scenario performs best, with its pollution reductions clustering most strongly in the city's southern neighborhoods, which carry the highest baseline air quality burden. The “high demand” scenario, however, outperforms equity-focused approaches by up to 57% in total energy savings, illustrating the efficiency-equity tension frequently discussed by urban planners. This benefit

evaluation system offers a structured way to move beyond simple utilization metrics and assess the full downstream impact of infrastructure deployment decisions.

#### Long-Term Governance Models:

The governance side of this transition is addressed by Pardo-Bosch et al. (2021), who analyze the charging network strategies of Nantes and Hamburg under the EU mySMARTLife project. Nantes pursued a maximum coverage strategy, installing stations across publicly accessible car parks funded through EU grants, municipal taxes, and private operator agreements. Hamburg focused on residential areas underserved by off-street parking, with a longer-term plan to transition management toward private operators as the market matured. Both cases illustrate what the authors call the “chicken-and-egg” dilemma: “the adoption of EVs depends on a robust network of public CSs, which in turn requires a considerable up-front investment with a long payback period that reduces the commercial viability for private organisations” (Pardo-Bosch et al., 2021). The Hamburg model reflects a theory of public infrastructure being a market catalyst by absorbing early-stage financial risk, allowing the city to create conditions under which private operators can eventually sustain the network. Pardo-Bosch et al. also emphasize that “there is a need to connect cities through electrified corridors as most people cannot afford to own different vehicles for urban and interurban travel” (Pardo-Bosch et al., 2021), pointing toward regionally integrated networks rather than municipal efforts that exist in isolation.

Taken together, these studies sketch out a future in which conventional parking infrastructure is gradually displaced by curbside charging zones, smart municipal hubs, and vehicle fleets that double as distributed energy storage. But even with this sketch of the future, the theoretical pathways toward it are varied. Some prioritize utilization and economic efficiency;

others center environmental justice and equitable access; and still others focus on the energy grid transformations that make the whole project viable. What the literature consistently makes clear is that the pace and shape of this transition will be determined less by the technology itself and more by the deliberate planning decisions cities make in the years ahead.

### ***Ethical and Legal Considerations:***

Yet even with the right infrastructure in place, there are major ethical considerations that must be addressed in order to scale AV adoption. Some people believe AVs should default to traditional normative ethical approaches, while others push for more progressive and dynamic ones. In this section, we will critique four different ethical frameworks: Utilitarianism, Deontology, Ethical Pluralism, and Ethical Rawlsianism to analyze which approach society should consider implementing.

#### **Utilitarianism (Consequentialism):**

Among the four approaches mentioned above, perhaps the most well known is a form of consequentialism called utilitarianism. To put it simply, utilitarianism aims to maximize the total benefit for the greatest number of people (Arfini et al., 721). This idea is best illustrated by the infamous trolley car problem, where a high speed train must decide between two alternatives: either continue on route and kill five people or swerve and kill only one person. At a surface level, the answer to save five individuals over one seems obvious and intuitive, but what happens when the demographic profile of the individuals change? What if the scenario instead involves five elderly people and one small child? Or one murderer and five doctors? This thought experiment has become increasingly more relevant in the context of AV ethical programming.

To best explain a major limitation of utilitarianism in the context of AV programming, philosophy professor Selene Arfini considers the following scenario: Suppose an AV encounters a road obstruction and must choose between the following options: kill a motorcyclist without a helmet, kill a motorcyclist with a protective helmet, or kill the passenger (Arfini et al., 718). Since utilitarianism aims to minimize harm, it will choose to hit the motorcyclist with the helmet. This choice, however, is largely problematic as it harms the person who chose to invest in their physical safety. This, as Arfini highlights, creates warped incentives for killing the most protected, and oftentimes most innocent, person on the road (Arfini et al., 718). Additionally, computing the utilitarian calculus for every possible outcome is not only infeasible but also raises legal and ethical concerns. This means the AV would have to assign a numerical value to every life involved and pursue the option that it deems less valuable. Not to mention engineers would have to physically encode these values into the car.

#### Deontology:

Deontology dates back to the 18th century and concerns itself with rules, laws, and duties. The ethical framework suggests that adhering to pre-defined rules is more important than the outcome itself. In the context of the trolley problem, the deontological approach would either always swerve or never swerve; always protect the passenger or always protect the innocent pedestrian. Deontology, however, raises some major limitations.

Most importantly, Arfini highlights that no set deontological rulebook can possibly cover every driving situation (Arfini et al., 723). Rules like “do not discriminate between individuals” would essentially mean “tossing a coin” to establish who to harm in a no-win scenario. Furthermore, rules like “always protect the passenger” would hinder widespread AV adoption, since certain neighborhoods would be reluctant to allow them on their streets (Arfini et al., 723).

### Moral Pluralism:

Perhaps the most promising ethical framework for AV adoption are those that consider multiple ethical theories at once. Field Safety Architect at Waymo and Stanford Mechanical Engineering Professor Chris Gerdes proposes an interesting blend between Consequentialism (Utilitarianism) and Deontology. He argues that the inherent nature of consequentialism to weigh outcomes and minimize harm closely maps to a math optimization problem where the car optimizes against these “costs” (Gerdes & Thornton, 91). However, deontology acts as constraints against the objective by enforcing hard coded rules. Stated another way, deontology provides the guardrails for the objective function to find a best alternative within those constraints. As a result, consequentialism defines the objective function and deontology roughly defines the constraints for the optimization problem (Gerdes & Thornton).

Similarly, Arfini et al., rejects both pure utilitarianism and deontology to instead propose a naturalist blend. They propose a three tiered plan that preserves national, company, and individual decision-making processes (Arfini et al., 731). First, they propose a United Nations equivalent for AV regulation. This committee would provide the deontological skeleton that acknowledges how different legislations and cultural perspectives can manifest in different AV laws. Second, OEMs could develop their own ethical policies within the parameters set by the national regulatory agency. Third, individuals can choose insurance policies that best align with their ethical views. Perhaps one OEM places more emphasis on either passenger safety or pedestrian safety. This ensures that people still have choice autonomy within a moderated environment (Arfini et al., 731).

### Rawlsian Approach:



Authors Claudia Brändle and Michael Schmidt highlight society's inherent moral incompatibility. Choosing a politically acceptable solution cannot refer solely to one ethical doctrine, as doing so would exclude other perspectives. As a result, Brändle and Schmidt propose Ethical Rawlsianism. Rawlsianism aims to find shared moral agreements (overlapping consensus) across different ethical frameworks to then make decisions based on this reflective equilibrium (Brändle & Schmidt, 1475). Essentially, AVs should be programmed to abide by constraints that manifest in both physical road laws and social contracts that would generally be agreed upon through a form of "public reason" (Brändle and Schmidt, 1495).

***Conclusion:***

Autonomous vehicles are among the most defining technological advancements of the 21st century. Although this period of accelerating innovation presents exciting possibilities, it also bears several economic, infrastructural, and ethical challenges. Ultimately, the success of autonomous integration depends on thorough planning, cross-company cooperation, and regulatory oversight, not just technological progress.